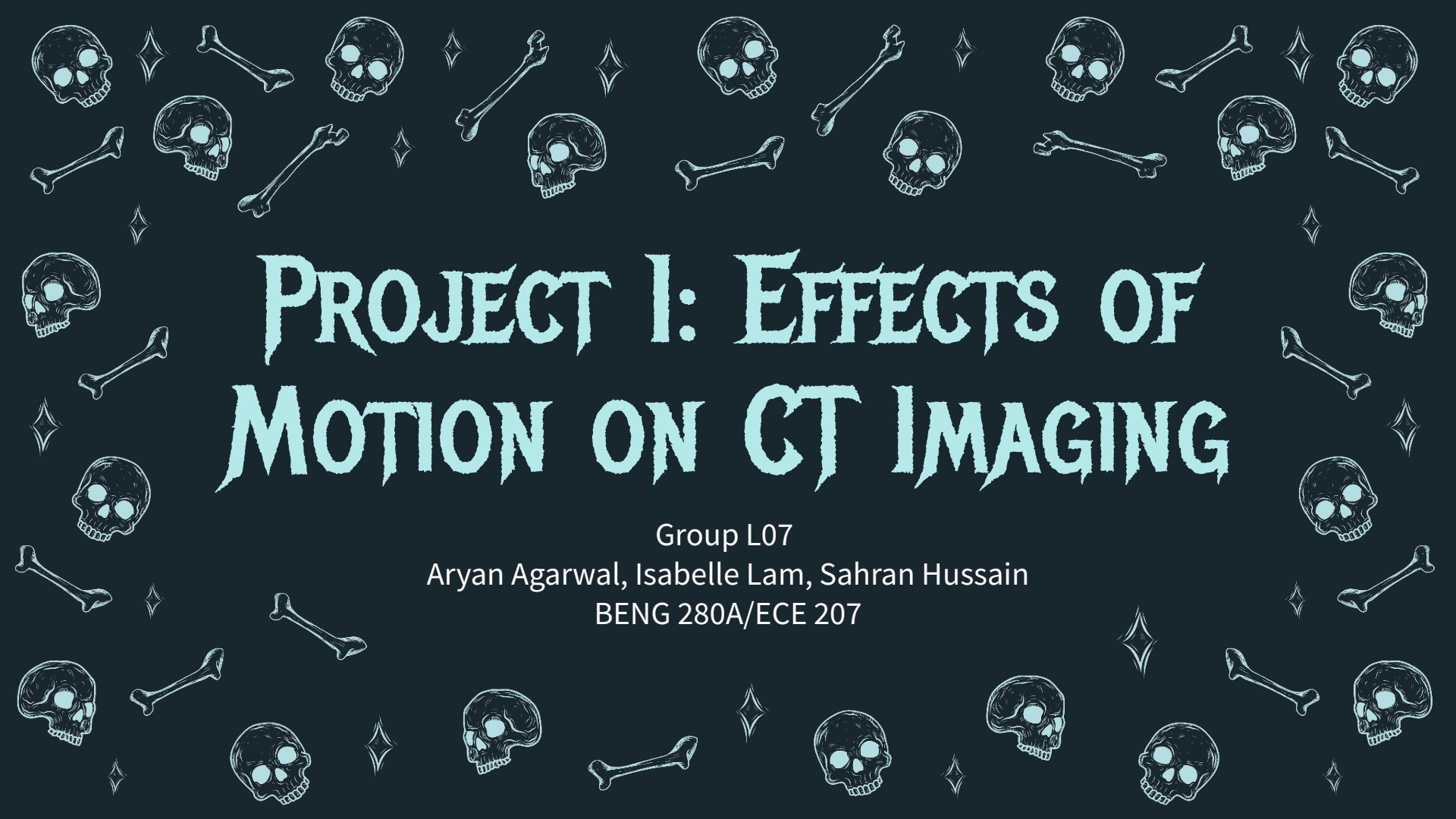


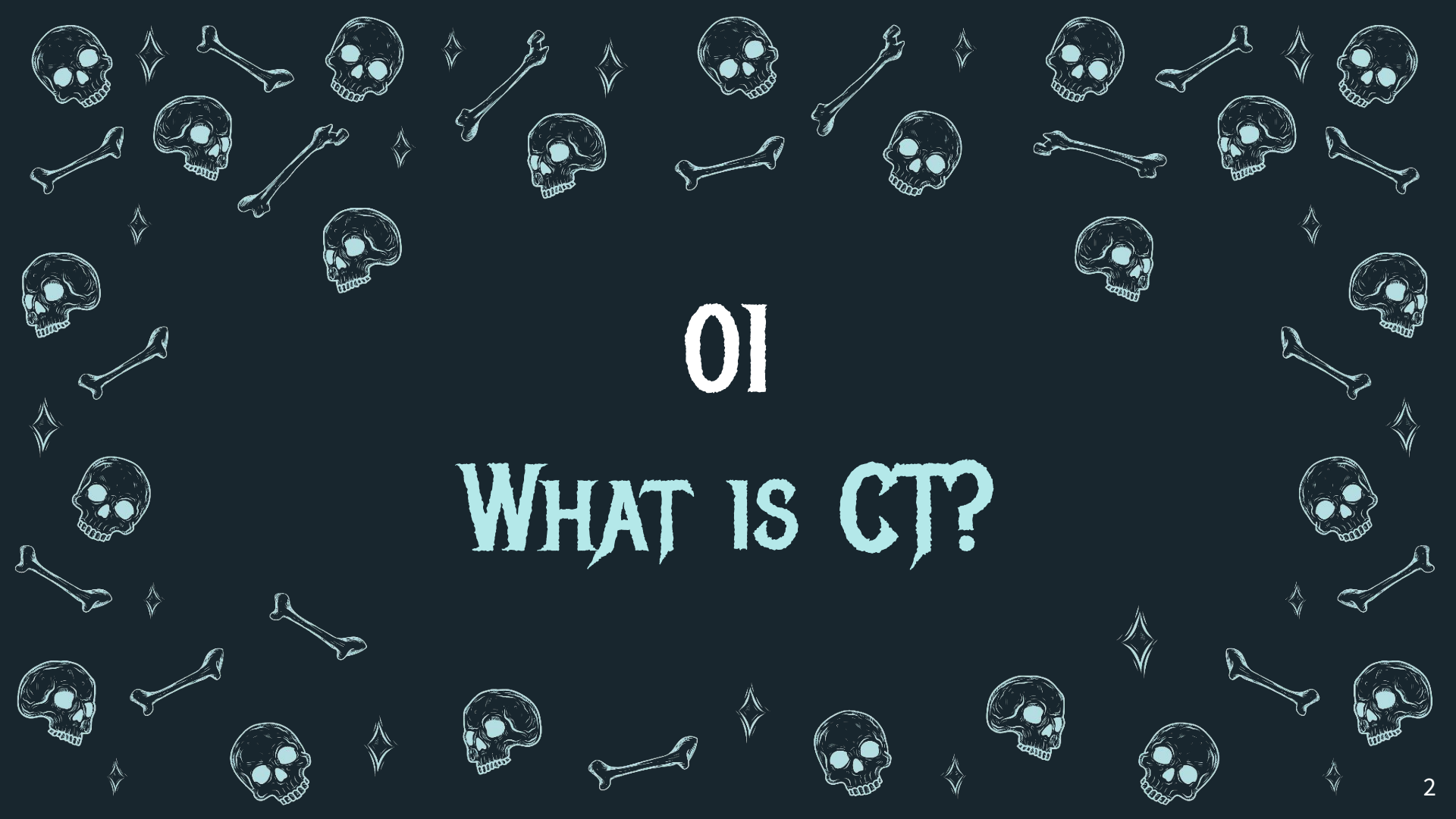
A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background. The skulls are shown from various angles, some with glowing eyes. The bones are long and thin, resembling femurs. The diamonds are small and white.

PROJECT I: EFFECTS OF MOTION ON CT IMAGING

Group L07

Aryan Agarwal, Isabelle Lam, Sahran Hussain

BENG 280A/ECE 207

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

01

WHAT IS CT?

COMPUTED TOMOGRAPHY IMAGING

- Computed tomography (CT) is an x-ray imaging technique using narrow beams rotating around a patient
- Generates “slices” that are 2D cross-sectional images stacked digitally to generate a 3D image
- CT scanners use a motorized x-ray source that rotates around “gantry” and are typically different from conventional x-ray imaging (x-ray tube)
- CT also use digital x-ray detectors, in contrast to typical x-ray film
- Patient lies on a platform that moves in and out of the gantry to collect images slice by slice [1]



Figure 1. Example CT [2]

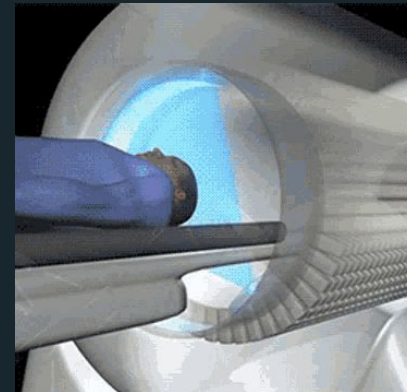
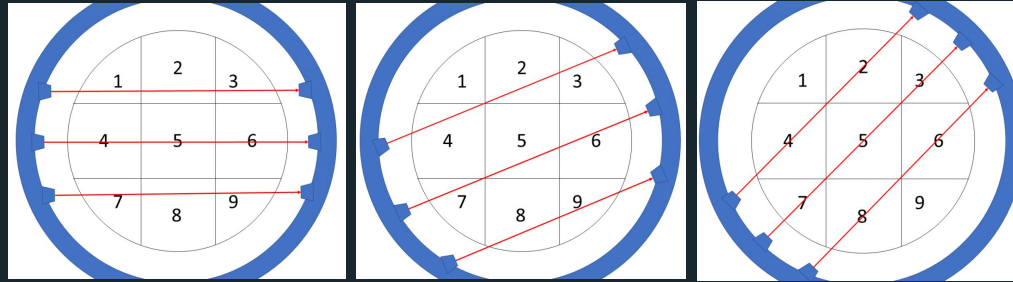
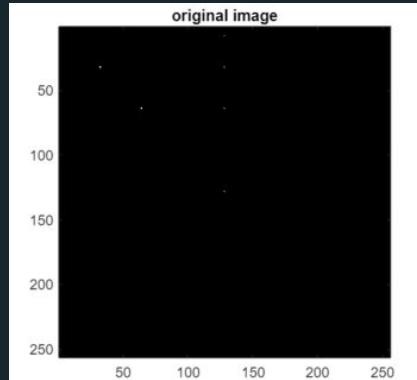


Figure 2. CT imaging process [3]

BASICS OF CT

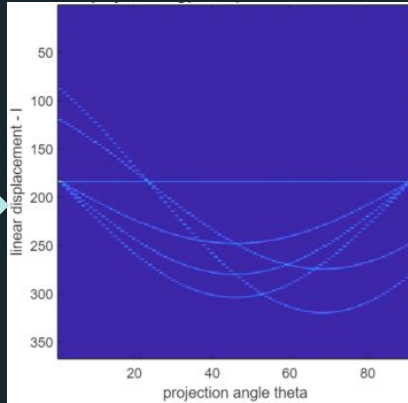


Ex HW #3



Radon

SINOGRAM



Inverse Radon

CT RECONSTRUCTION

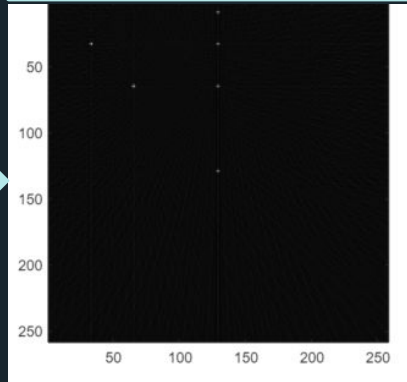


Figure 3. Breakdown of CT reconstruction [4][5]

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

02

LITERATURE REVIEW

TYPES OF MOTION IN CT IMAGES

- Literature identifies 3 common types of motion that arise in CT images
 - Cardiovascular and Peristalsis (involuntary): contractile and expansion → streaking and high contrast regions
 - Respiratory (involuntary): Z-axis blurring → loss of contrast
 - Muscle (voluntary & involuntary): Off-axis shift, aperiodic

[6][7][8]



Figure 4. *CT images of heart* [9]

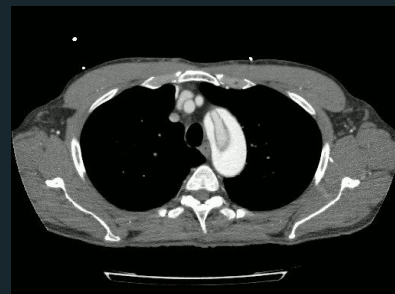


Figure 5. *CT images of lungs* [10]

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

03

CONTROL, LINEAR MOVEMENT, AND VIBRATION

TRANSFORMATIONS DURING MOTION

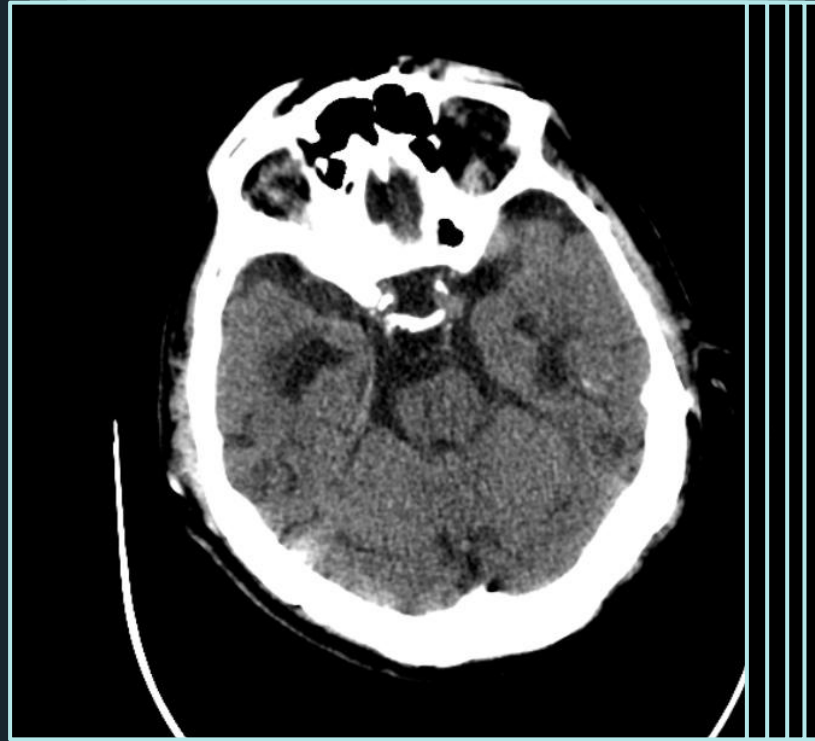


Figure 6. Axial motion Case CT [11]

Case courtesy of Dr David Cuete, Radiopaedia.org, rID: 25637

TRANSFORMATIONS DURING MOTION

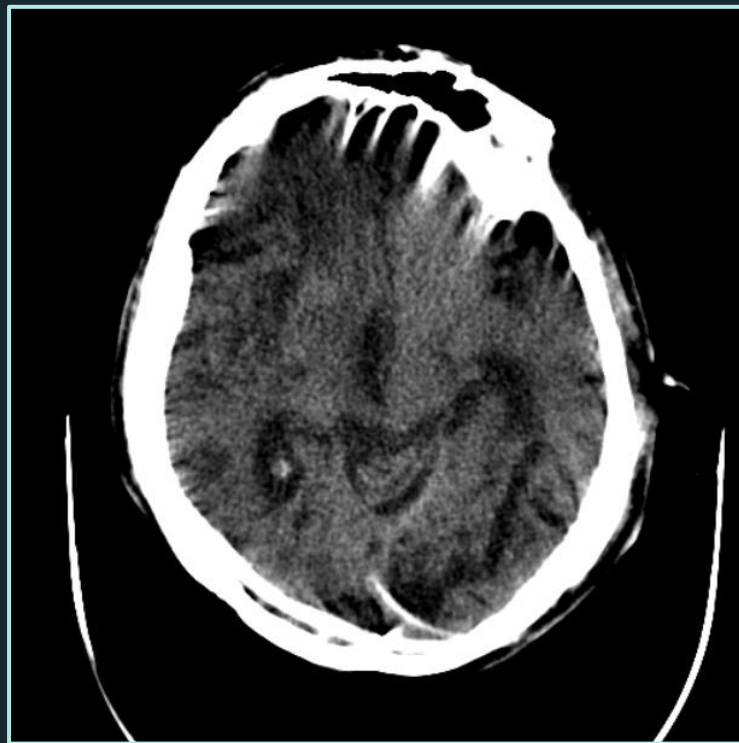
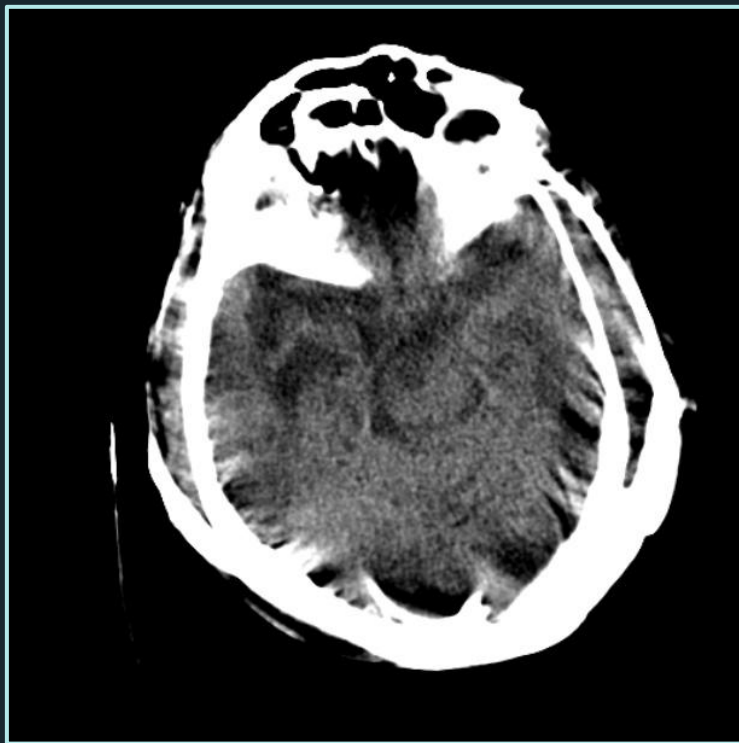
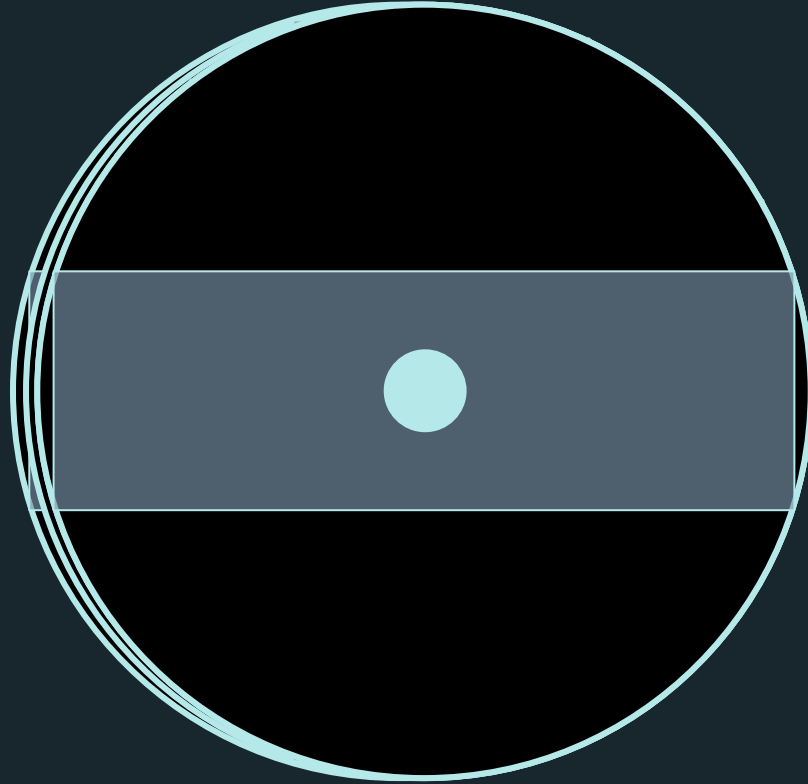


Figure 6. Axial motion Case CT [11]

Case courtesy of Dr David Cuete, Radiopaedia.org, rID: 25637

METHODOLOGY – TRANSFORMATIONS DURING MOTION



SAMPLE CODE BREAKDOWN - TRANSFORMATIONS DURING MOTION

%for loop to create the control image of a non moving circle

for i=1:10

x= -128:128;

% defining x for meshgrid

z= -128:128;

% defining y space for meshgrid

[x,z] = meshgrid(x,z);

% creating meshgrid

J = zeros(size(x));

% pre-allocating J matrix for generating images

J((x.^2+z.^2)<=(10)^2)=1;

% Creating Circle control image

I(:, :, i)=J;

% adding each J matrix to an element of the original I matrix to the 3rd dimension

% generating figures for each movement step based on n matrix index

figure('Name','original control image','Position',[1 420 400 400]);

imagesc(I(:, :, i)); title('original control image'); axis('square'); colormap('gray');

F = getframe(gcf);

%generating the gif file for the movement

imwrite(rgb2gray(F.cdata), filename, 'gif', 'WriteMode', 'append', 'DelayTime', 0);

clf

angles = theta(count:count+17);

%defining range of angles for each projection

rad_I = radon(I(:, :, i), angles);

%radon transform of images

inv_I = inv_I + iradon(rad_I, angles);

%inverse radon transform for the images

inv_I_Hamming = inv_I + iradon(rad_I, angles, "hamming");

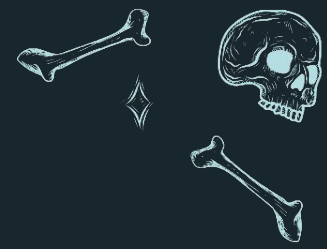
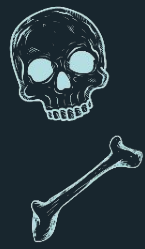
%inverse radon transform for hamming

count = count + 18;

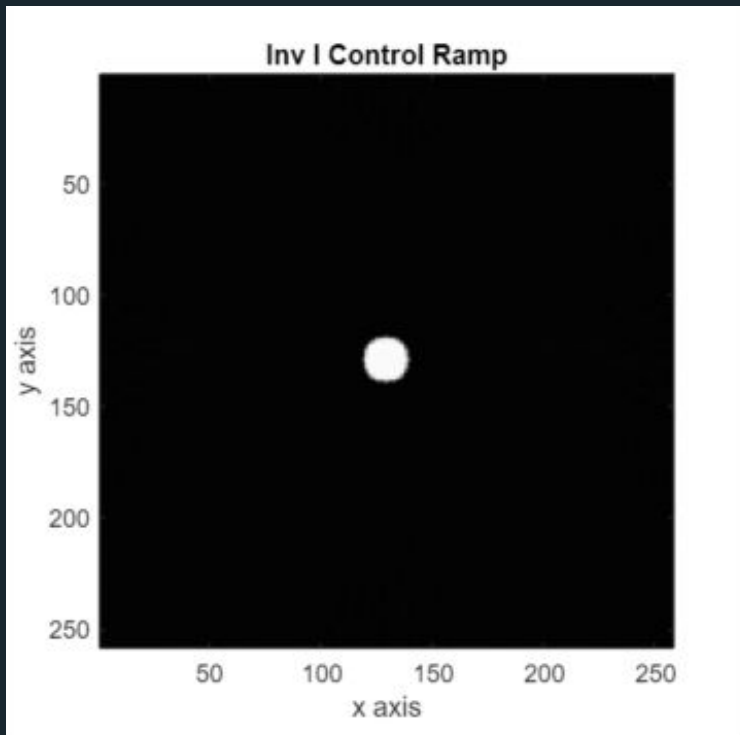
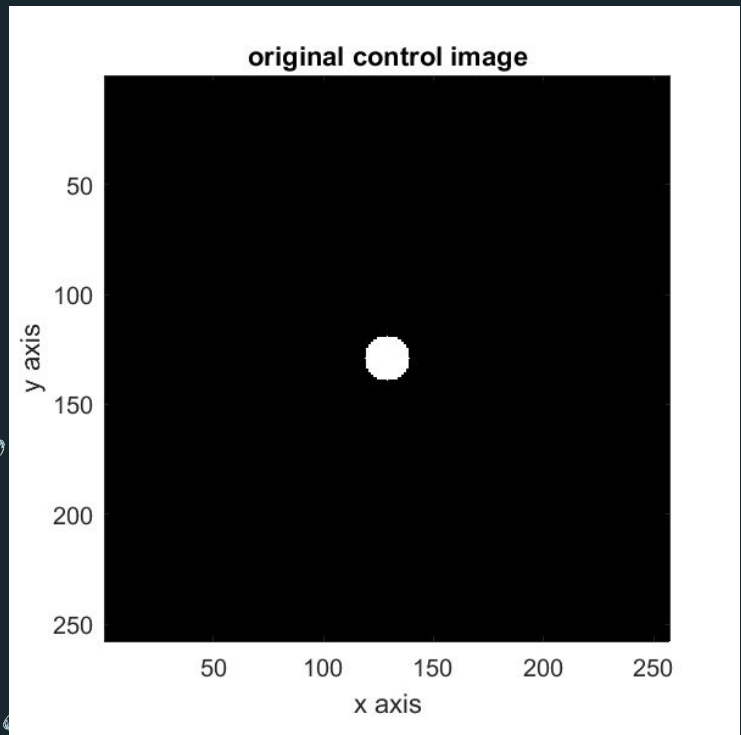
%changing count with each iteration for projection angles

end

CONTROL: No movement



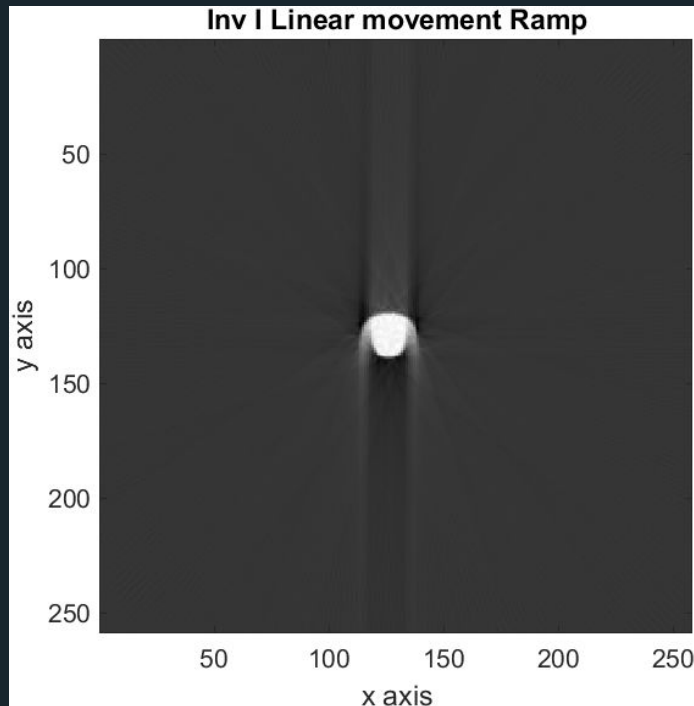
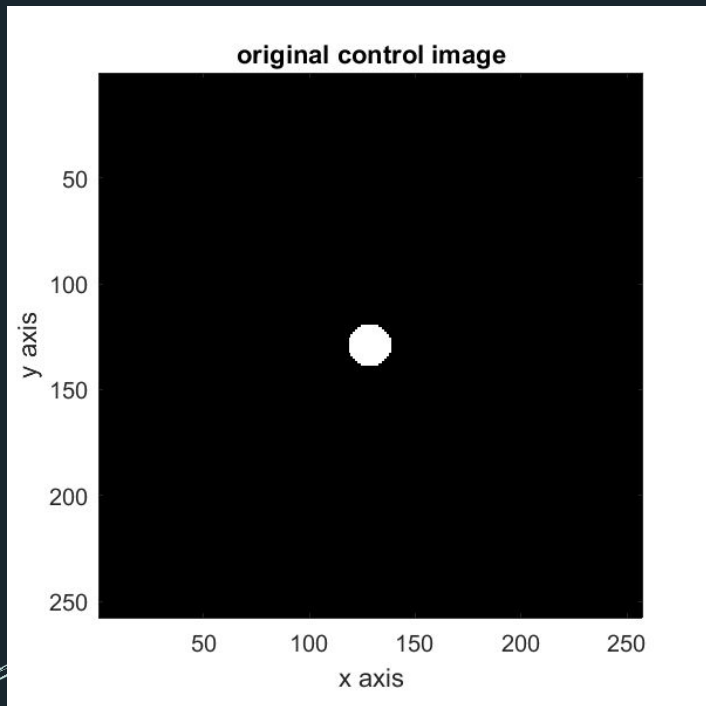
RESULTS - CONTROL



SNR = 396

RESULTS -

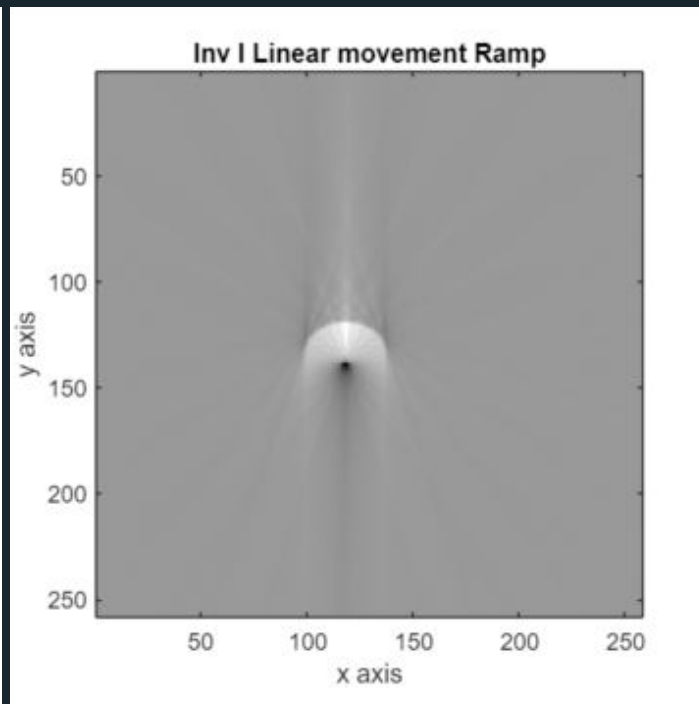
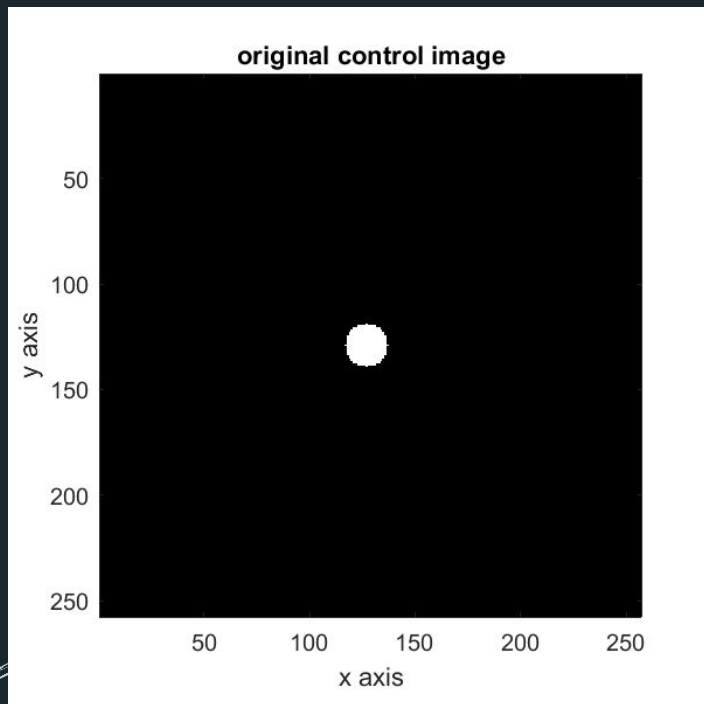
LINEAR MOVEMENT (0.5 PIXELS/IMAGE)



SNR = 276

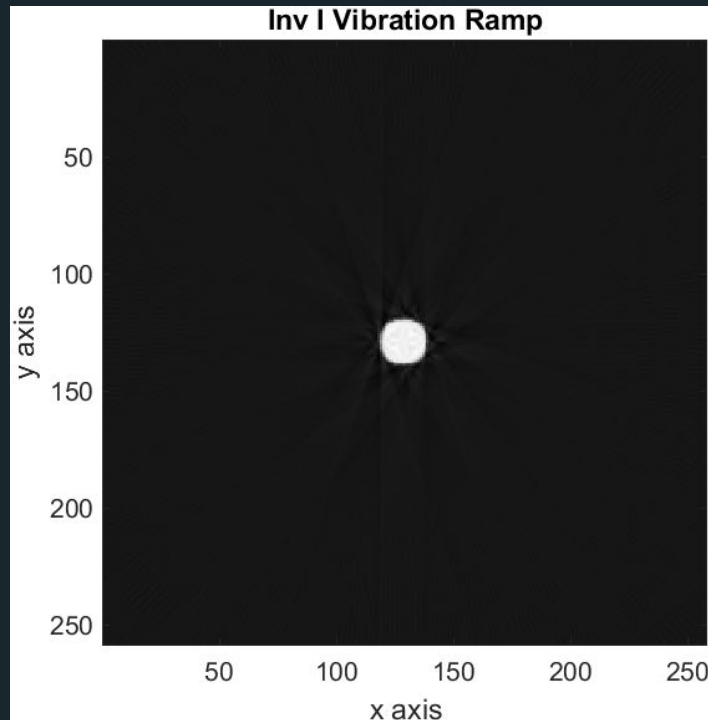
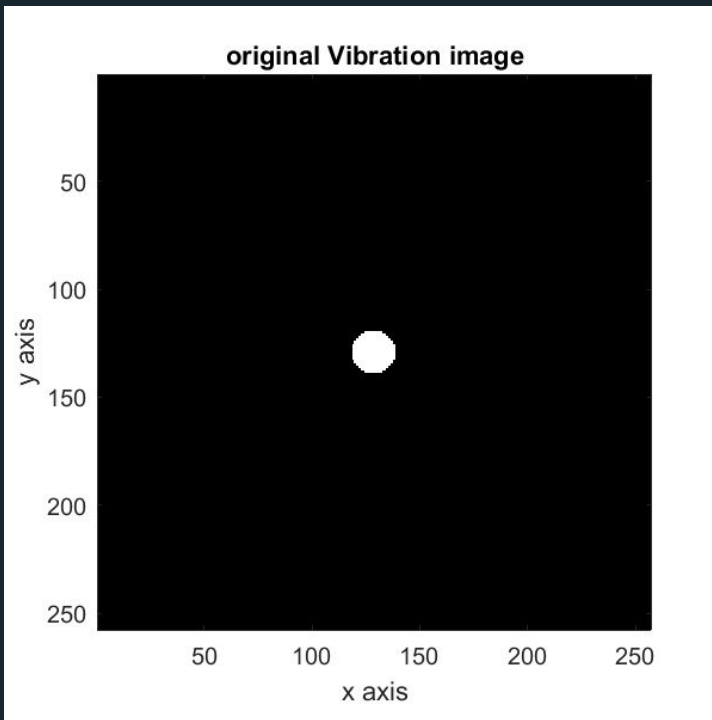
RESULTS -

LINEAR MOVEMENT (2 PIXELS/IMAGE)



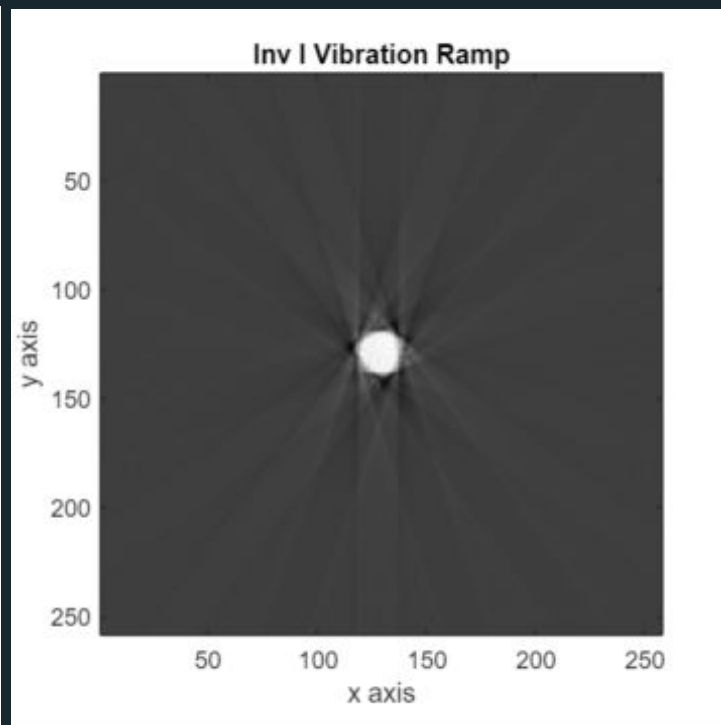
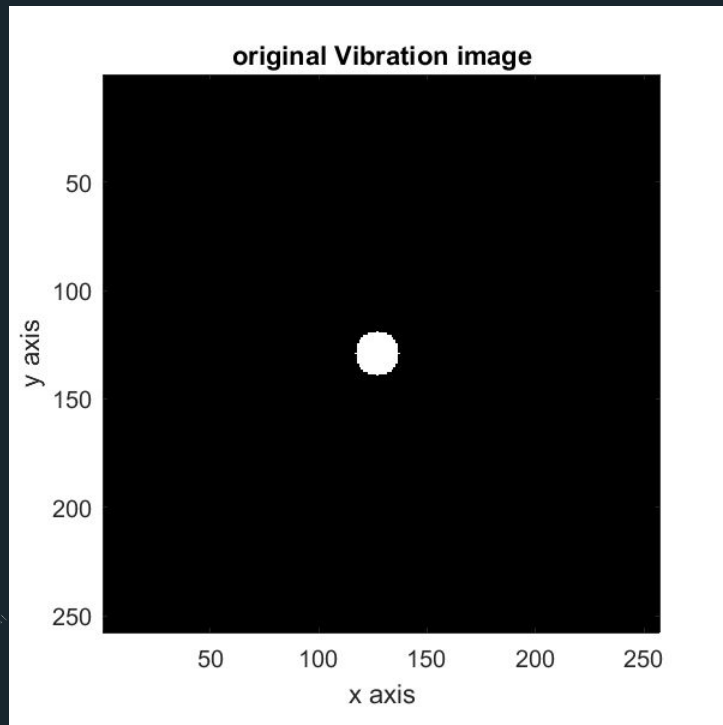
SNR = 130

RESULTS - VIBRATION (0.5 PIXELS/IMAGE)



SNR = 255

RESULTS - VIBRATION (2 PIXELS/IMAGE)



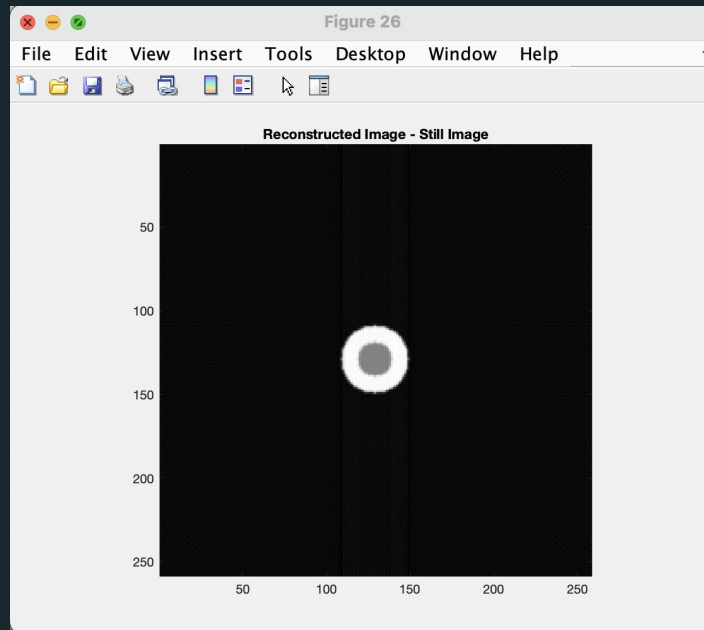
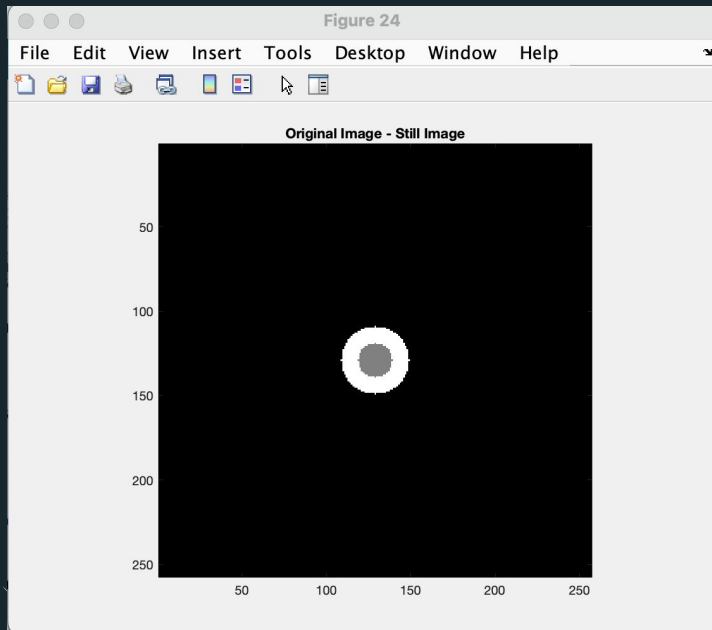
SNR = 146

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

04

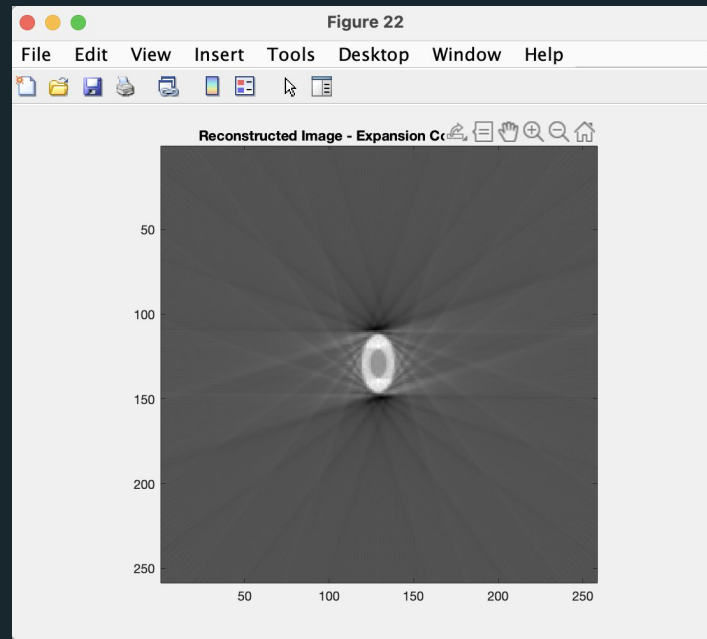
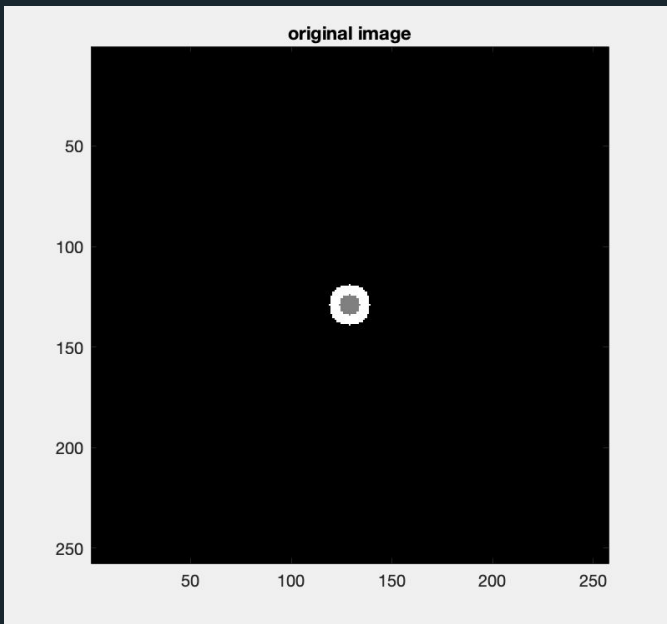
CONTRACTILE AND EXPANSION BLUR

RESULTS - CONTROL



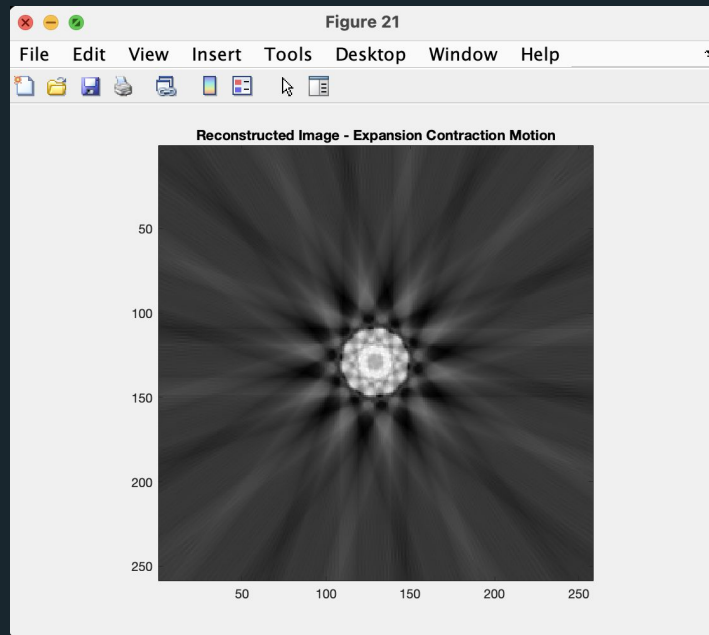
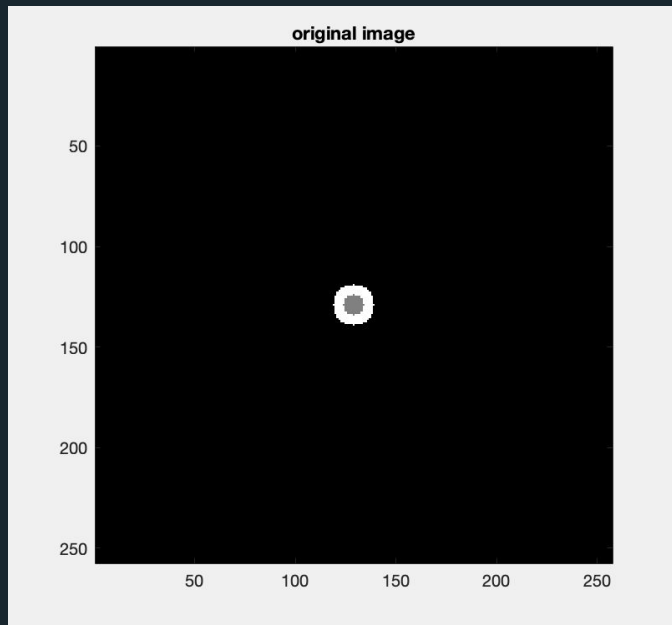
SNR = 203

RESULTS - GRADUAL CONTRACTION AND EXPANSION BLUR



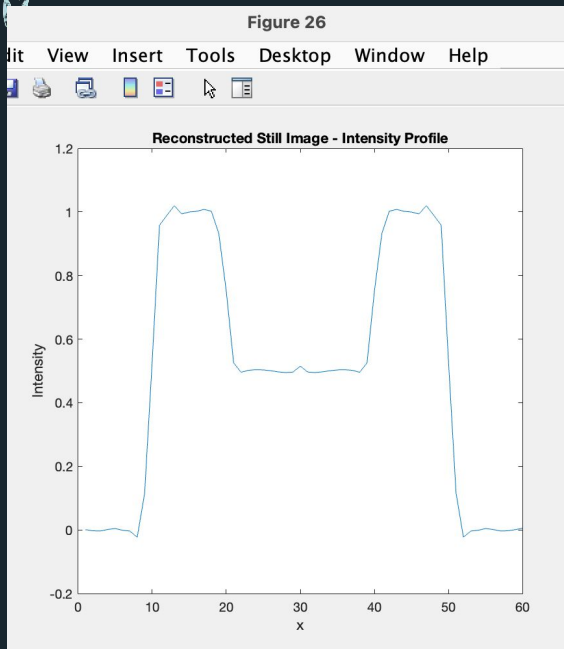
SNR = 45

RESULTS - RAPID CONTRACTION AND EXPANSION BLUR

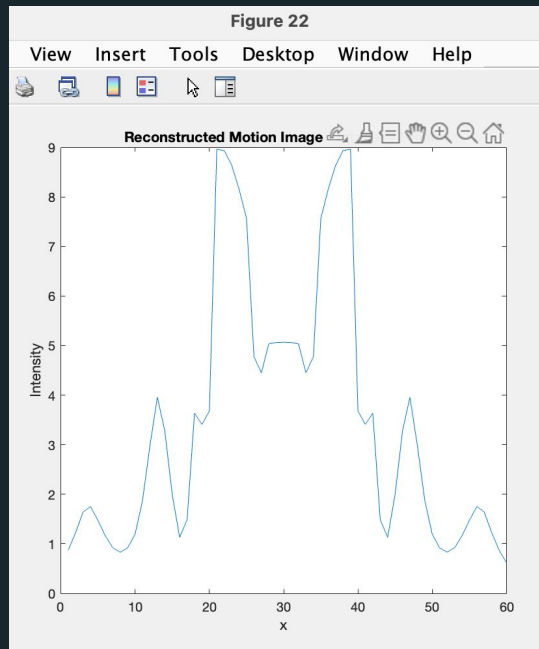


SNR = 19

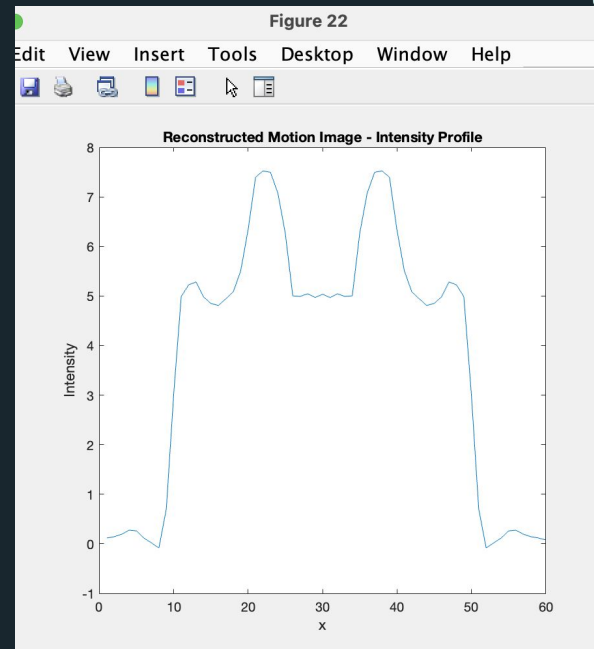
RESULTS - INTENSITY PROFILES



Control



Gradual



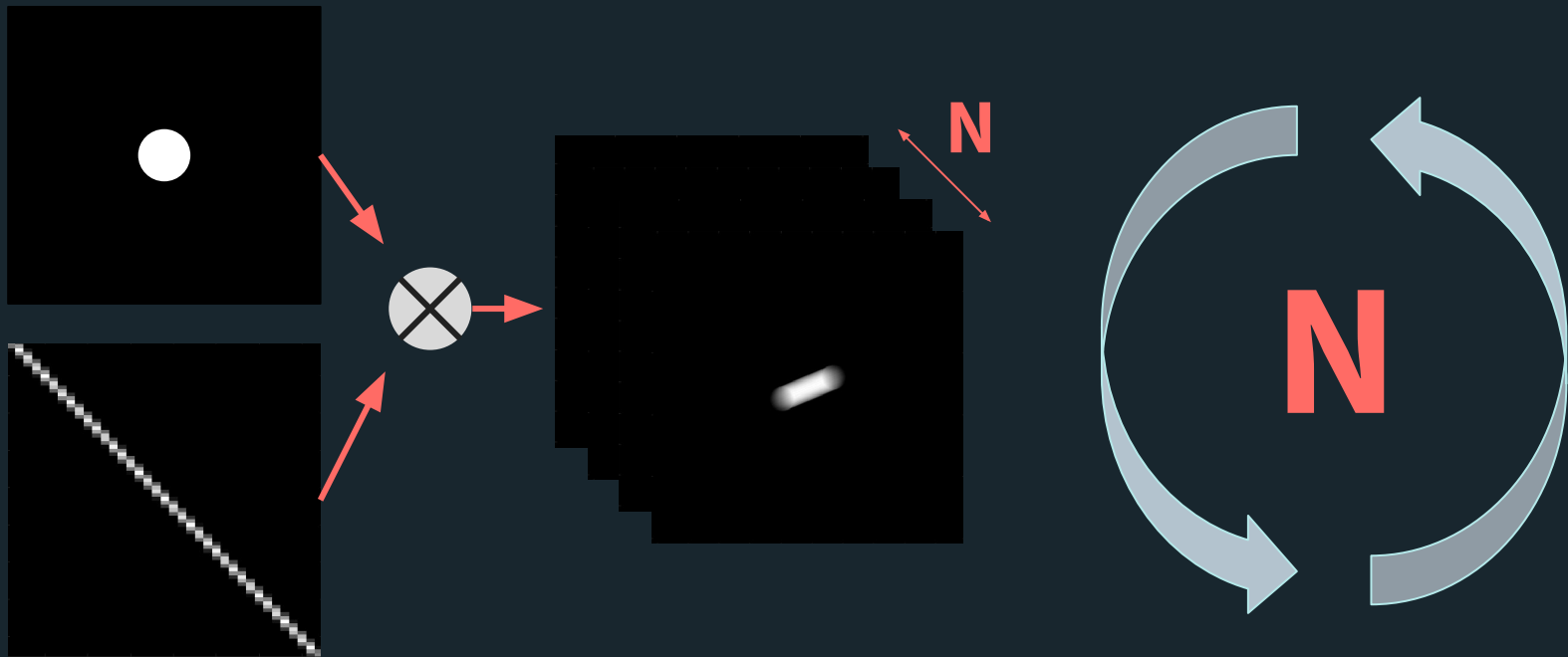
Rapid

A decorative border surrounds the central text, featuring stylized white skulls, bones, and four-pointed stars (diamonds) on a black background.

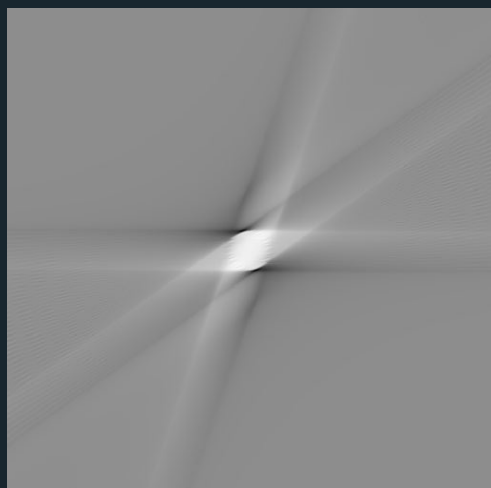
05

MOTION BLUR

MOTION BLUR SIMULATIONS

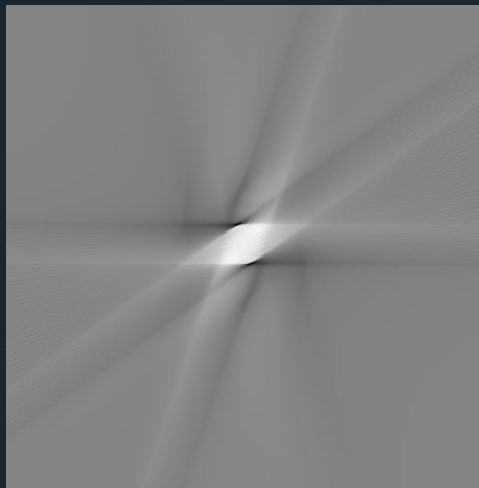


MOTION BLUR SIMULATIONS



1

+



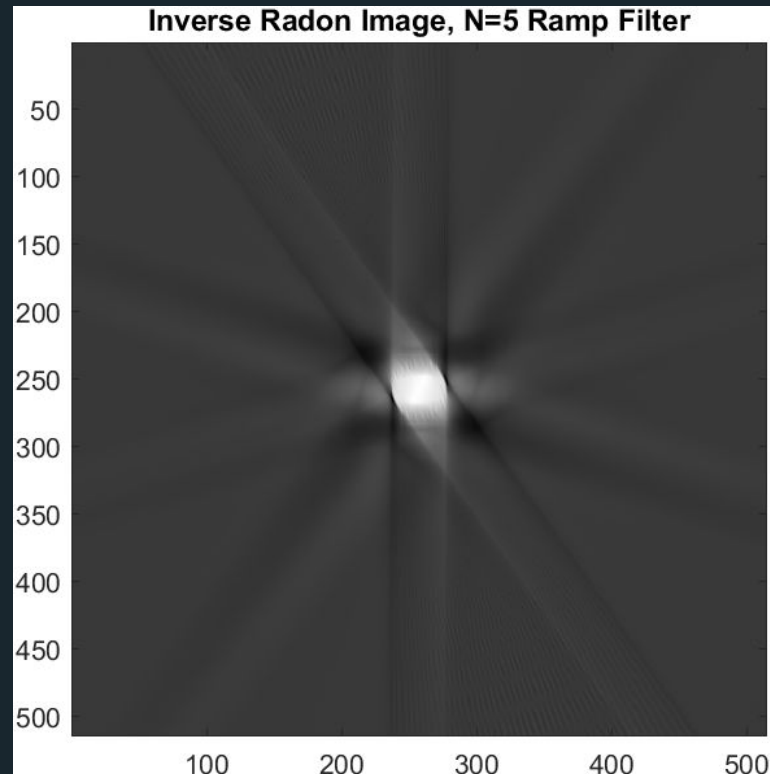
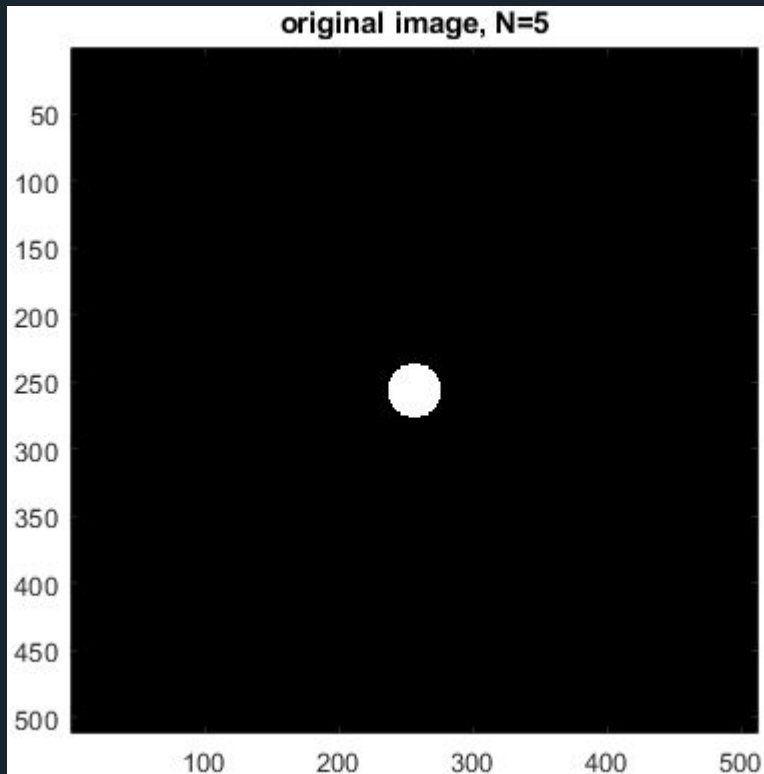
2



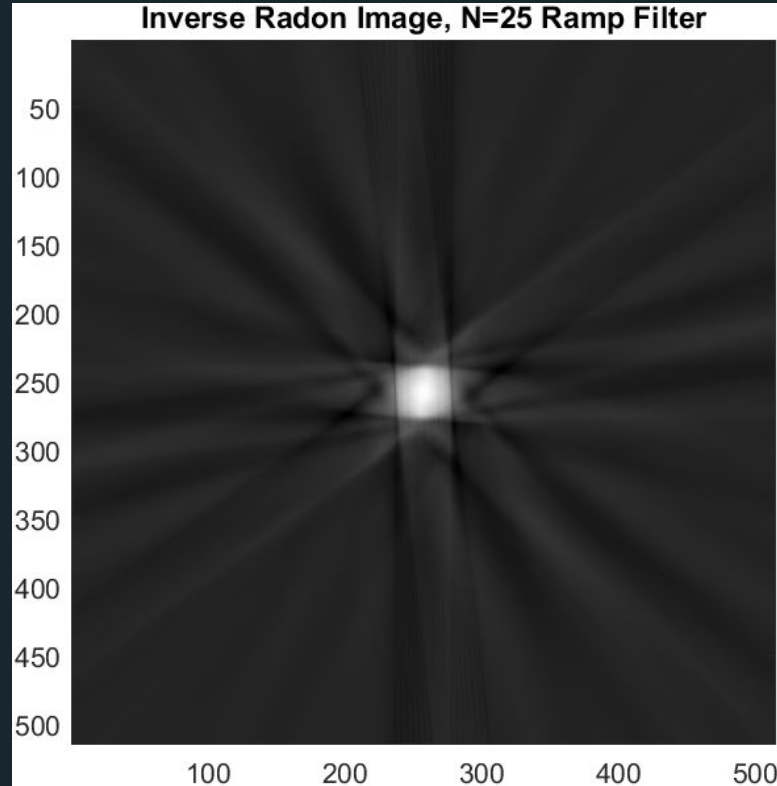
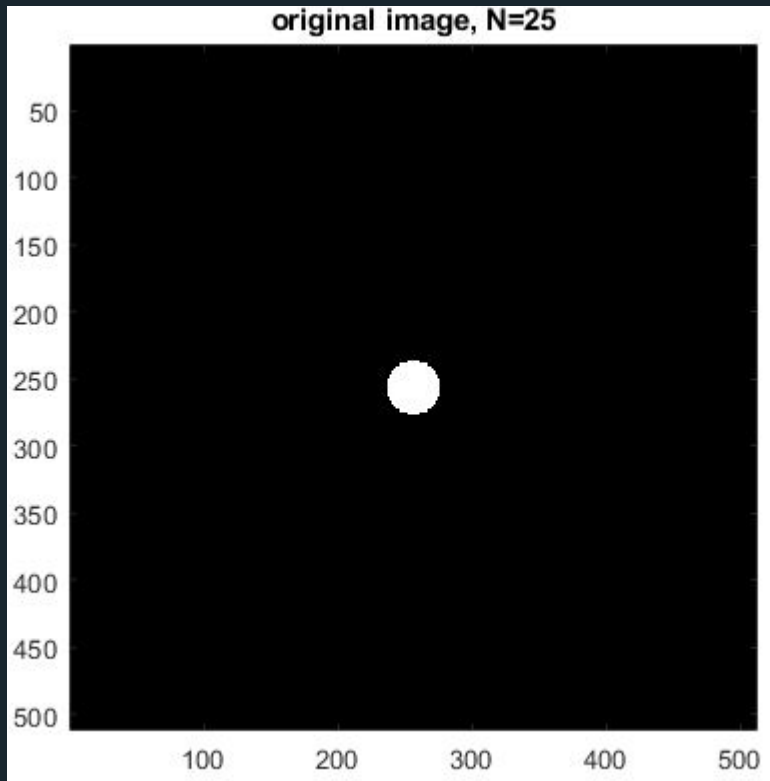
FINAL
RADON
IMAGE

N

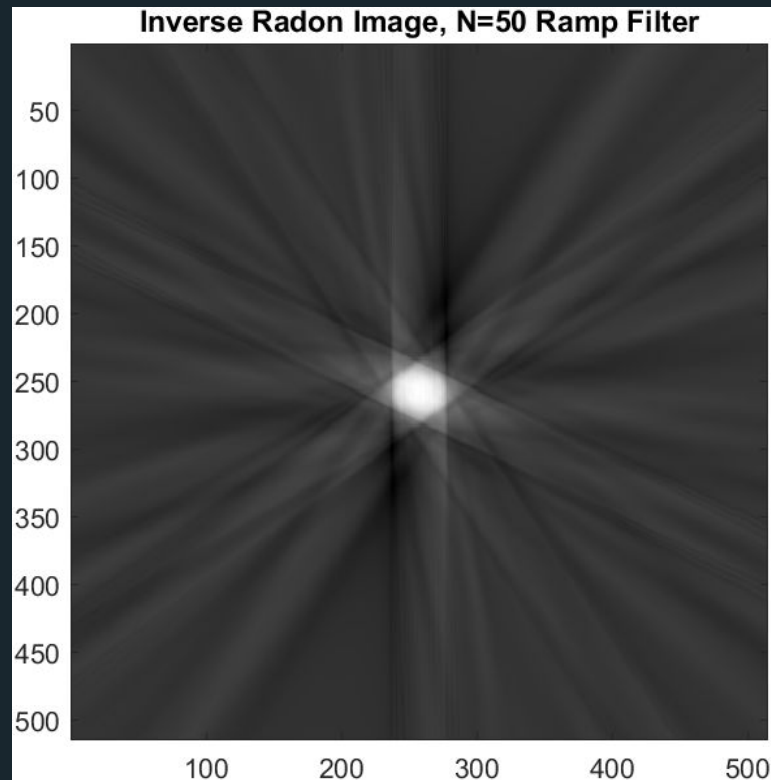
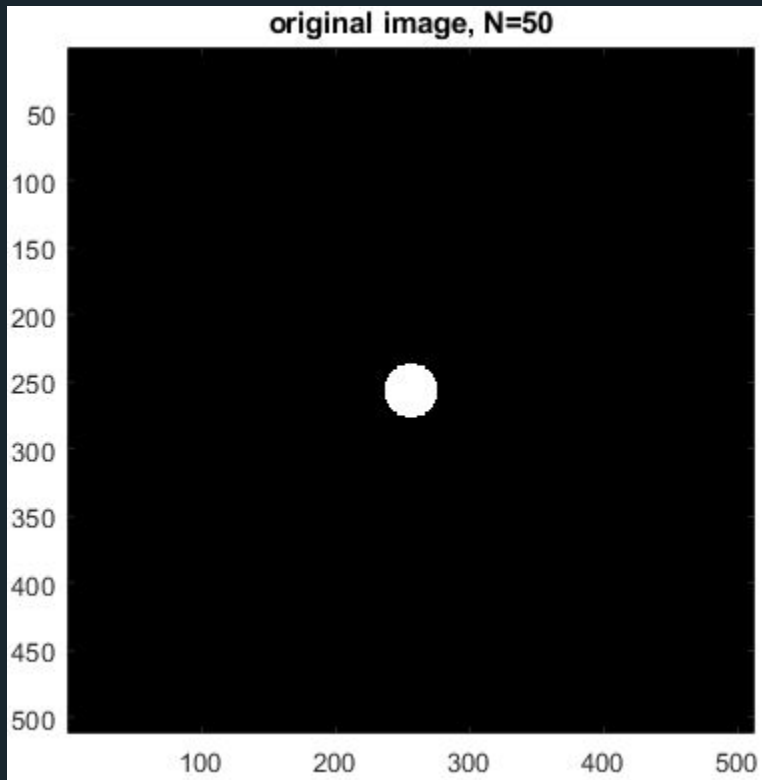
MOTION BLUR IMAGES



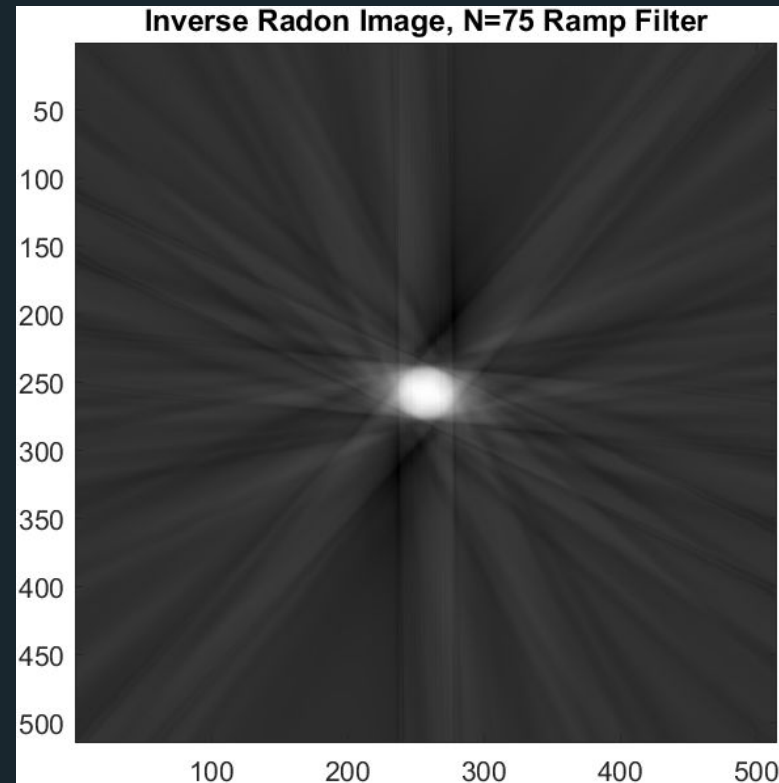
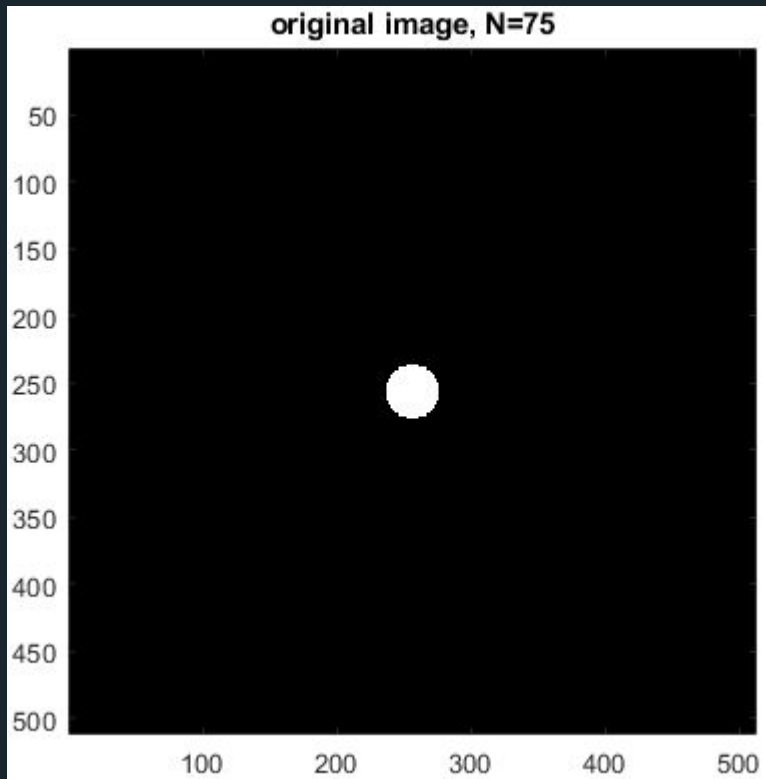
MOTION BLUR IMAGES



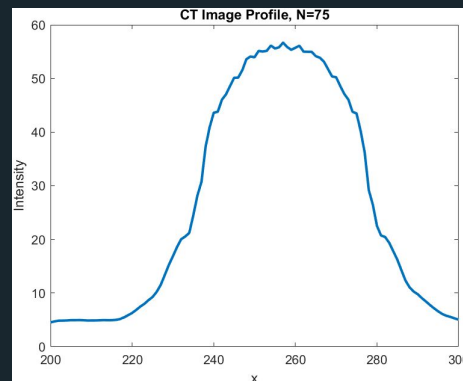
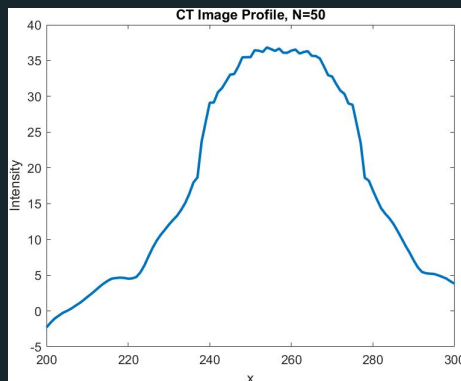
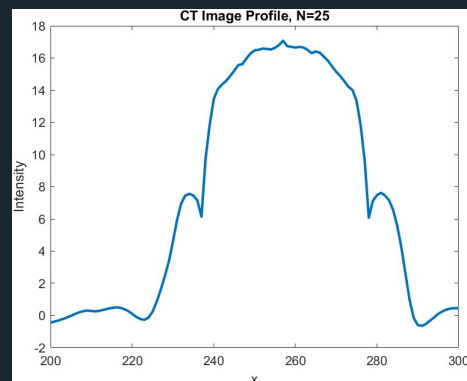
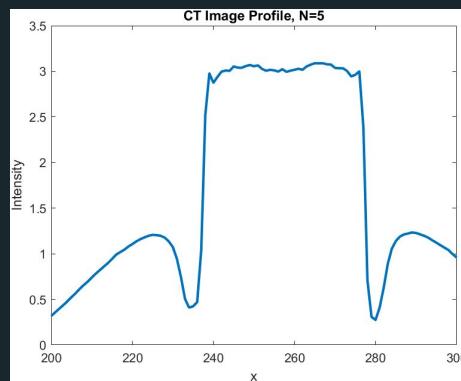
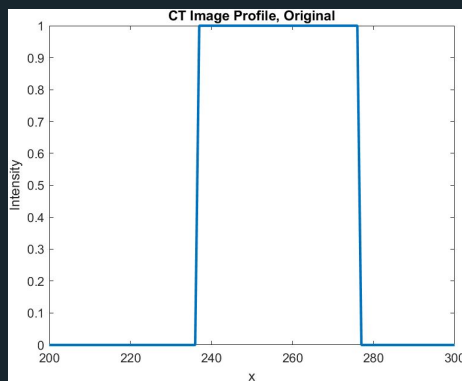
MOTION BLUR IMAGES



MOTION BLUR IMAGES



PRECISION OF MOTION BLUR



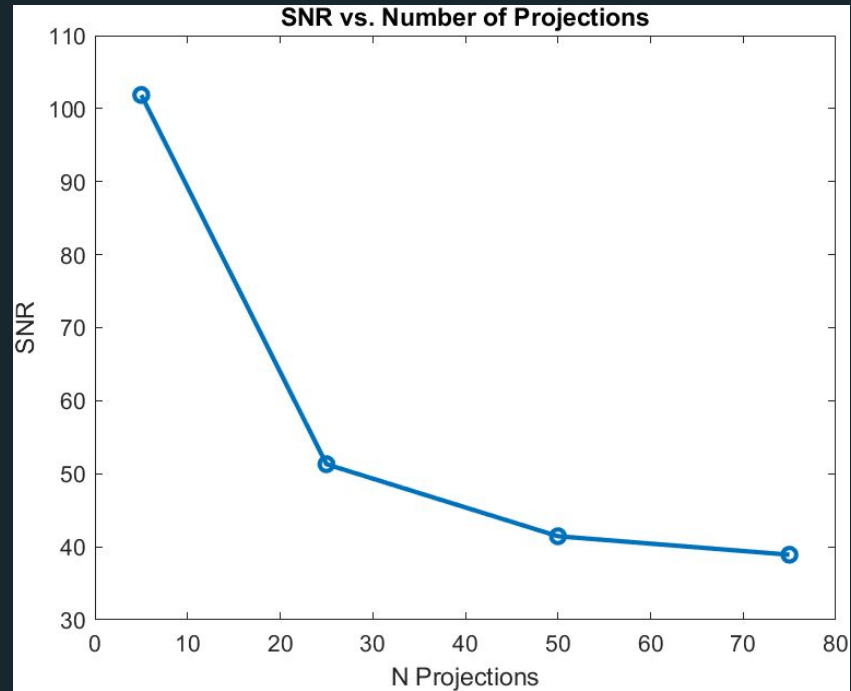
QUALITY OF MOTION BLUR

$$\text{SNR} = \frac{\{\max(I) - \text{mean}(B)\}}{\text{std}(B)}$$

I = signal

B = background

Control (no blur) SNR = 396





CONCLUSIONS AND FUTURE PROSPECTS

- Linear Movement and Vibration: increasing movement and rate of movement negatively affect image quality
- Contraction and Expansion blur: increasing rate of movement negatively impacts image quality
- Motion blur: compromise between quality and precise imaging

Future

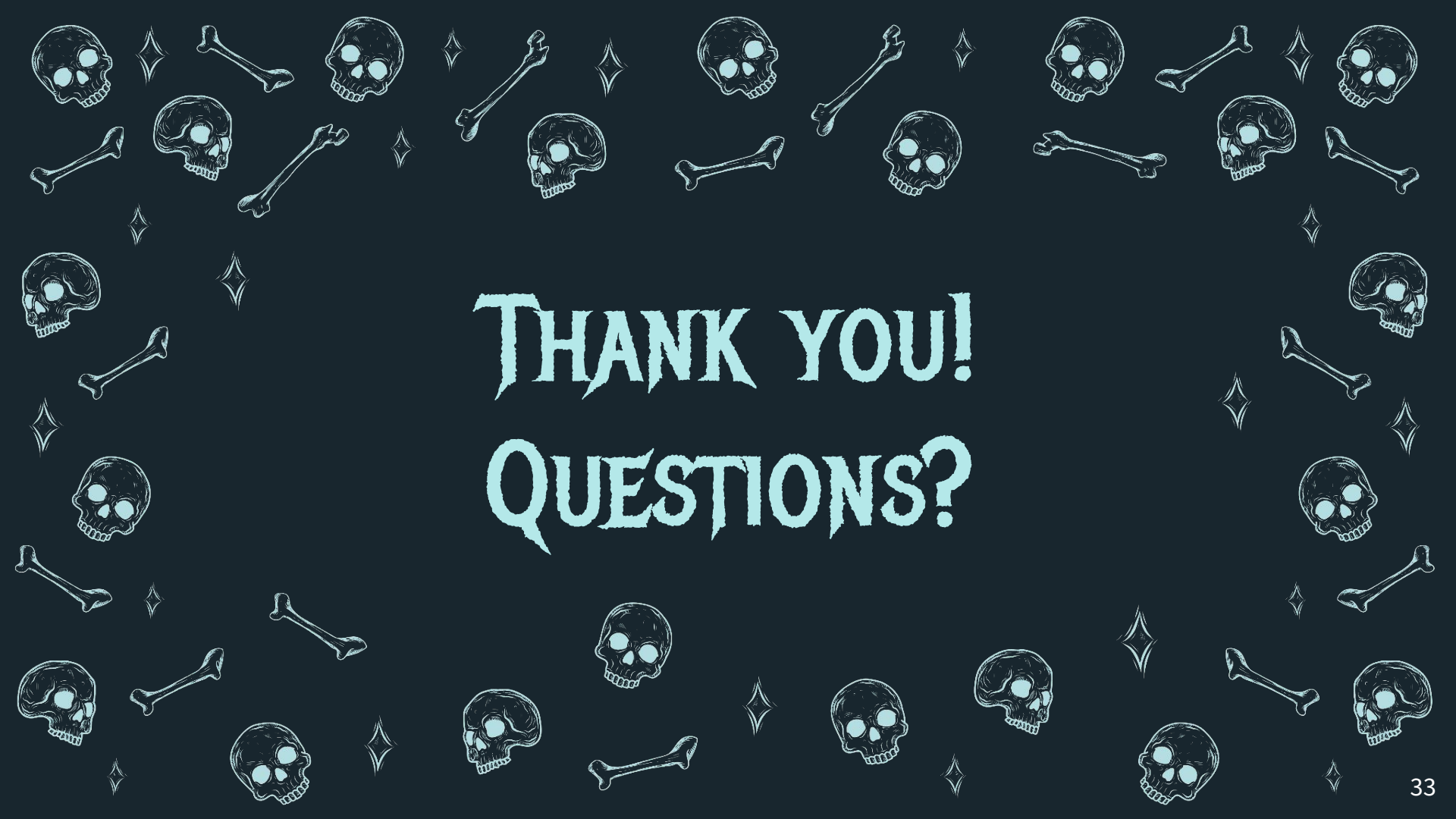
- Simulation on real patient data to see effects
 - Implementing reconstruction filter algorithms to remove motion artifacts
 - AI/Machine Learning to predict what an image should look like when disturbed by motion artifacts
- 



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THANK YOU!
QUESTIONS?